

The Chain Rule

MATH 145 – Applied Calculus

Torin Quinlivan

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Composition of Functions

Recall when we talked about composition of functions:

Composition of Functions

If f and g are functions such that $g : A \rightarrow B$ and $f : B \rightarrow C$, we define the **composition of f and g** to be the new function $h : A \rightarrow C$ give by

$$h(t) = f(g(t)).$$

What if we wanted to find the derivative of h ? Is there a good way to do this, if we know f' and g' but not h' ?

Example: Composition of Functions

Example: For the following expressions, which is the “outer function”, f and which is the “inner function”, g ?

■ $(5x^4 + 3x^7)^9.$

■ $\sqrt{3x^2 + 2x + 1}.$

■ $\frac{2}{x^3 + x^6}.$

■ $\sin(x^2).$

The Chain Rule

Composition of Functions

If f and g are differentiable functions, then:

$$\frac{d}{dx}(f \circ g)(x) = \frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x).$$

Proof of the Chain Rule

$$\begin{aligned}\frac{d}{dx}f(g(x)) &= \lim_{h \rightarrow 0} \left[\frac{f(g(x+h)) - f(g(x))}{h} \right] \\&= \lim_{h \rightarrow 0} \left[\left(\frac{f(g(x+h)) - f(g(x))}{h} \right) \left(\frac{g(x+h) - g(x)}{g(x+h) - g(x)} \right) \right] \\&= \lim_{h \rightarrow 0} \left[\left(\frac{f(g(x+h)) - f(g(x))}{g(x+h) - g(x)} \right) \left(\frac{g(x+h) - g(x)}{h} \right) \right] \\&= \lim_{h \rightarrow 0} \left[\frac{f(g(x+h)) - f(g(x))}{g(x+h) - g(x)} \right] \times \lim_{h \rightarrow 0} \left[\frac{g(x+h) - g(x)}{h} \right] \\&= f'(g(x)) \times g'(x)\end{aligned}$$

Example: The Chain Rule

Example: Find the derivative of the following functions:

■ $f(x) = (5x^4 + 3x^7)^9.$

■ $g(x) = \sqrt{3x^2 + 2x + 1}.$

■ $h(x) = \left(x^4 + \frac{3}{x}\right)^3 + \sqrt{x}.$

■ $q(x) = \frac{2}{\sqrt{x^2 + 3x}}.$

■ $\ell(x) = \sqrt[3]{(x^2 + 4x)(x^5 + 7x^3)}.$

Example: The Chain Rule

Example: If $f(x) = (4x^3 + 5x^2)^8$, find $f'(x)$ and $f''(x)$.

Example: The Chain Rule

Example: If $f(x, y) = \sqrt{x^2 + xy + y^2}$, find $\frac{\partial f}{\partial x}$.